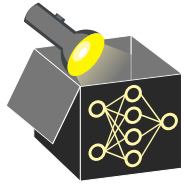
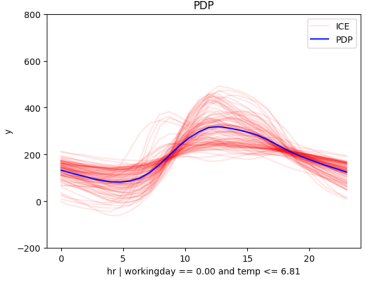
Interpretable Machine Learning



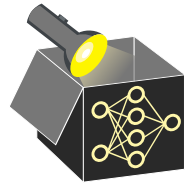
Feature Effects
Further Resources and Software



Learning goals

- Learn about further resources
- Get an overview of software packages in R and Python

RESOURCES AND SOFTWARE:



Individual Conditional Expectation (ICE)

- The paper: "Goldstein et al." 2015
- Chapter in Interpretable Machine Learning book: "Molnar" 2025
- R packages: "iml" n.d.; "pdp" n.d.; "ICEbox" n.d.
- Python packages: "PiML" n.d.; "sklearn" n.d.

Partial Dependence (PD) plot

- Chapter in Interpretable Machine Learning book: "Molnar" 2025
- R packages: "iml" n.d.; "pdp" n.d.; "DALEX" n.d.
- Python packages: "sklearn" n.d.; "effector" n.d.; "PDPbox" n.d.

Accumulated Local Effect (ALE)

- The paper: "Apley et al." 2020
- Chapter in Interpretable Machine Learning book: "Molnar" 2025
- R packages: "ALEPlot" n.d.; "iml" n.d.
- Python packages: "effector" n.d.; "PiML" n.d.; "ALEPython" n.d.; "Alibi" n.d.

Regional effect plots (REPID)

- The paper: "Herbinger et al." 2022
- Codebase: "repid" n.d.